

MARCO CHEN

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EDUCATION

University of Waterloo

Bachelor of Computer Science, Co-op

Sep. 2023 – Aug. 2028

CGPA: 3.7/4.0

- **Coursework:** OOP, DSA, Operating System, Compiler, Graph Theory, Probability, Statistics, Calc (1-3), LinAlg (1-2)

EXPERIENCE

Machine Learning Enginner

WAT.ai | Python, PyTorch, CUDA, WandB, SoccerNet

June 2026 – Current

Waterloo, ON, Canada

- Researching on World Model for soccer dynamics, building a Selective SSM + Imagined Rollout + Actor-Critic architecture to predict next latent state and planning strategic actions.

Software Engineer Intern

Geotab | C#, TypeScript, PostgreSQL, Google BigQuery, React, .NET

May 2026 – Aug. 2026

Kitchener, ON, Canada

- Optimized harsh driving event recommendation load latency by parallelizing API calls and refactored slow database queries, improved vehicle address reverse geocoding API latency by 90% and total request latency by 75%.
- Implemented a data persistence layer and an offline-first synchronization module using a local message queue for a telematics mobile app, caching user activities and auto-syncing payloads upon network reconnection, enabling full offline access and cutting redundant API calls by 40%.

Software Engineer Intern

Geotab | C#, TypeScript, PostgreSQL, React, .NET

Sep. 2025 – Dec. 2025

Kitchener, ON, Canada

- Optimized and refactored service initialization and synchronization logic, reducing JSON-RPC API calls by 46%.
- Led the automation of the Transports Canada ELD test procedures, resulting in 98% reduction in execution time.
- Investigated and debugged backward compatibility bugs, improved consistency across different OS platforms.

Software Engineer Intern

Octopodi Technologies | Rust, Typescript, SQLite, React, Next.js, Tauri

Jan. 2025 – April 2025

Waterloo, ON, Canada

- Developed a cross-platform desktop application using Tauri/Next.js to identify coding plagiarism, integrating advanced pattern-matching algorithms and customizable detection policies to improve analytics accuracy.

PROJECTS

Mini-vLLM | Python, PyTorch, MPS, KV Cache, PagedAttention

- Designed a minimal implementation of vLLM, optimized inference performance for Apple Silicon chips using MPS.
- Implemented optimization techniques including PagedAttention, Continuous Batching, Prefill-Decode Separation, Chunked-Prefill, Prefix Caching, and Tensor Parallelization for optimized inference throughput and TTFT.

LLM From Scratch | Python, PyTorch, CUDA, WandB, Multiprocessing, einops

- Implemented the full RMSNorm, SwiGLU FFN, scaled dot-product Multi-Head Attention with causal masking and RoPE on Q/K, achieving a validation perplexity of 2.20 and validation loss of 0.78 on the TinyStories dataset.
- Developed a training pipeline using BF16 mixed-precision, zero-copy strided batching, custom AdamW optimizer, cosine annealing, and gradient clipping, sustaining 413k tokens/sec throughput verified on an H100 GPU.
- Built an optimized BPE tokenizer with a max-heap and multiprocessing, reducing merge complexity to $O(n \log m)$.

LLM Gateway | Go, PostgreSQL/pgvector, Redis, Kafka, ClickHouse, Prometheus/Grafana, Docker, Kubernetes

- Architected a multi-tenant LLM proxy in Go handling routing, caching, rate-limiting, cost control, and observability.
- Engineered a two-tier semantic cache for vector embeddings, dropping tail latency from 1.4s to 470ms.
- Built custom net/http adapters and Redis sliding-window rate limiters for full streaming control, sustaining 1,200+ RPS with sub-8ms P99 overhead and <2ms check latency across 10K concurrent keys.

TECHNICAL SKILLS

Languages: Python, Go, TypeScript, C#, C++, Kotlin, Rust, Java, HTML, CSS, SQL

Frameworks: chi, FastAPI, LangGraph, Next.js, React, Django, Node.js, Flask, Tailwind CSS, Jest

Tools & Platforms: CUDA, Git, Linux, ClickHouse, Prometheus/Grafana, Docker, Kubernetes

Databases/Cloud: GCP, AWS, S3, Lambda, PostgreSQL, pgvector, MySQL, MongoDB, Redis, Kafka

Machine Learning: PyTorch, WandB, TensorFlow, Keras, OpenCV, NumPy, Matplotlib, pandas, PyAudio